

CS 5542 – Lab 9 Individual Contribution Report

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1. Personal Contributions

1.1 Railway Cloud Deployment

I set up and configured the Railway deployment for the entire application, creating two independent services connected to the GitHub repository with auto-deploy on push:

- Backend service — FastAPI running via uvicorn in a Docker container, root directory set to /backend
- Frontend service — Streamlit running in a Docker container, root directory set to /frontend
- Configured all environment variables in the Railway dashboard including Snowflake credentials, Gemini API key, and private key for authentication
- Resolved multiple deployment failures including Python detection issues, Docker image size limit (8.1 GB → 2.8 GB), and PORT expansion errors

1.2 Snowflake Key-Pair Authentication

The original system used Duo MFA which cannot run in a cloud deployment. I investigated and implemented an automated authentication solution:

- Attempted Programmatic Access Token (PAT) — blocked by EDU account network policy, confirmed by running SHOW GRANTS TO ROLE TRAINING_ROLE (no SECURITYADMIN/ACCOUNTADMIN privileges)
- Implemented RSA key-pair auth — generated 2048-bit RSA key pair using OpenSSL, registered public key with Snowflake user via ALTER USER, stored private key as Railway environment variable
- Updated scripts/sf_connect.py to detect SNOWFLAKE_PRIVATE_KEY and switch to snowflake_jwt authenticator automatically, preserving original MFA flow for local dev
- Verified live connection: CHEETAH → ROHAN_BLAKE_KENNETH_WH / CS5542_PROJECT_ROHAN_BLAKE_KENNETH

1.3 Docker Containerization

Wrote and debugged Dockerfiles for both services:

- Resolved 8.1 GB image size by installing CPU-only PyTorch before other requirements, staying within Railway's 4.0 GB free tier limit
- Fixed ModuleNotFoundError by aligning Docker working directory with uvicorn app path (backend.app:app from repo root)
- Fixed \$PORT not expanding by removing startCommand from railway.json and using \${PORT:-8080} in Dockerfile CMD
- Added build-essential system dependency for scispacy/spacy compilation on slim Python image

1.4 Frontend Environment Variable

Updated frontend/app.py to read BACKEND_URL from environment, defaulting to http://localhost:3001 for local development. Added BACKEND_URL to the frontend Railway service variables pointing to the live backend.

2. GitHub Commit Evidence

All commits are in the rohan/lab_9 branch:

https://github.com/BigDataAnalytics-CS5542/snowflake-research-assistant/tree/rohan/lab_9

3. Deployment URLs

Frontend (live application):

<https://frontend-production-c6c11.up.railway.app>

Backend API:

<https://backend-production-b66e.up.railway.app/health>

4. AI Tools Used

I used Claude (Anthropic) as an AI development assistant throughout this lab. Claude assisted with diagnosing Railway build failures, designing the key-pair authentication approach, debugging Docker image size and PORT issues, and explaining Railway's Nixpacks build system behavior. All generated code was reviewed, tested, and verified against live Railway logs before committing.